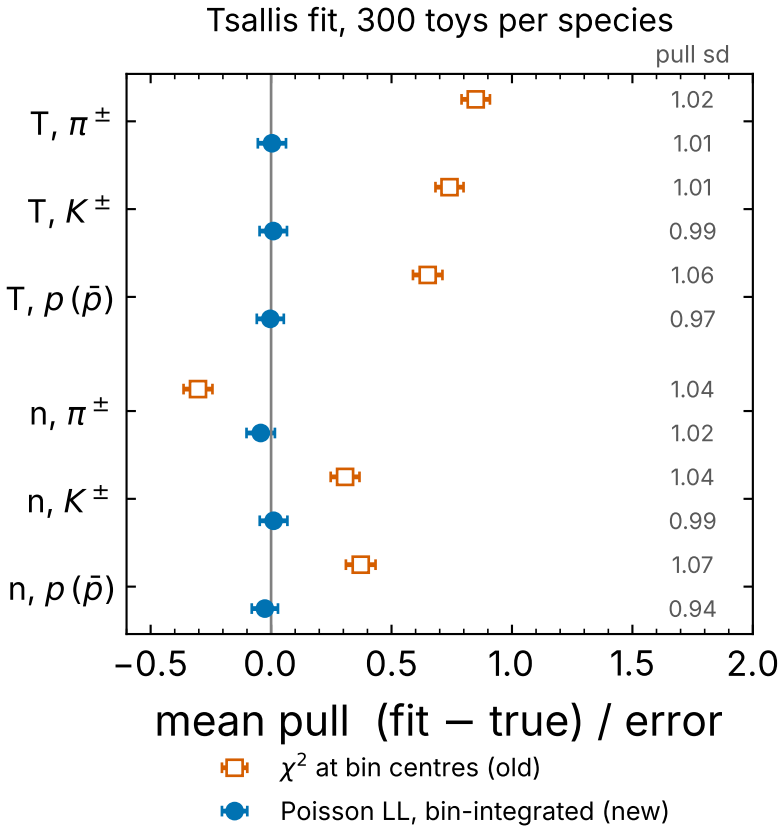
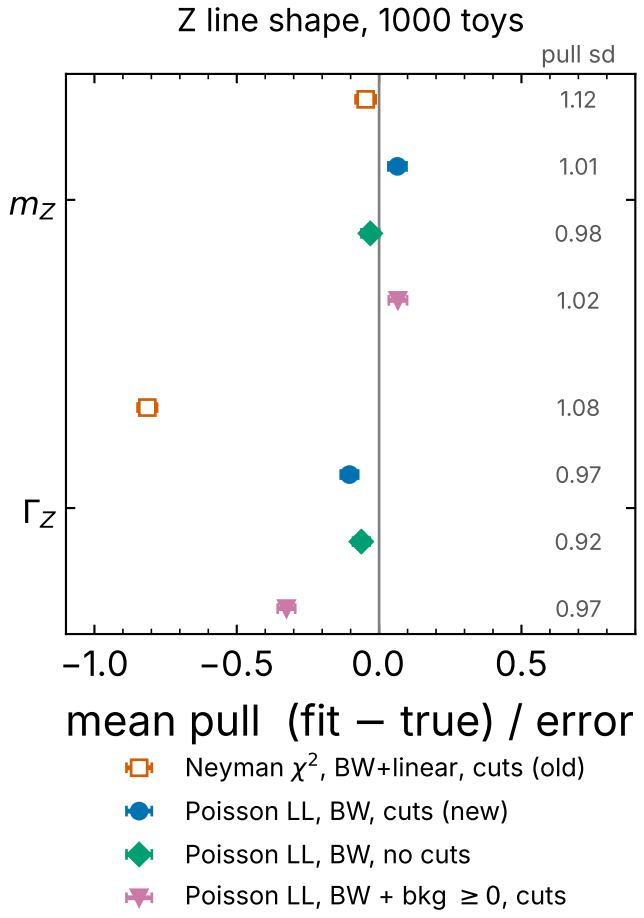
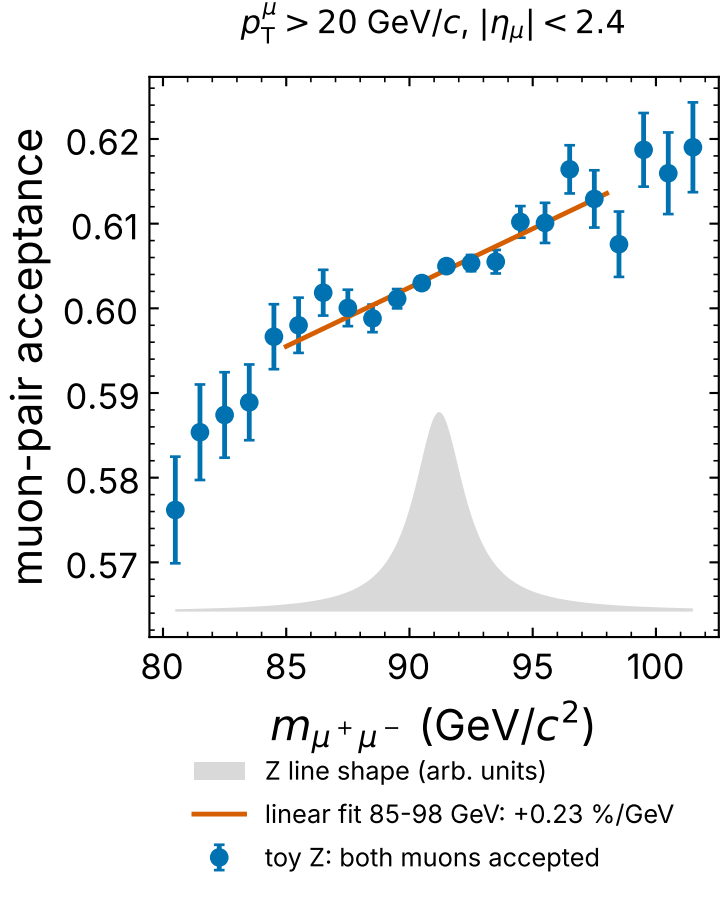

Tsallis fit, 300 toys per species
pull sd
T, $\pi^\pm$
T, $K^\pm$
T, $p\,(\bar{p})$
n, $\pi^\pm$
n, $K^\pm$
n, $p\,(\bar{p})$
1.02
1.01
1.01
0.99
1.06
0.97
1.04
1.02
1.04
0.99
1.07
0.94
−0.5
0.0
0.5
1.0
1.5
2.0
mean pull (fit − true) / error
χ^2 at bin centres (old)
Poisson LL, bin-integrated (new)


Z line shape, 1000 toys
pull sd
m_Z
Γ_Z
1.12
1.01
0.98
1.02
1.08
0.97
0.92
0.97
−1.0
−0.5
0.0
0.5
mean pull (fit − true) / error
Neyman χ^2, BW+linear, cuts (old)
Poisson LL, BW, cuts (new)
Poisson LL, BW, no cuts
Poisson LL, BW + bkg ≥ 0, cuts


$p_T^\mu > 20$ GeV/c, $|\eta_\mu| < 2.4$
muon-pair acceptance
0.62
0.61
0.60
0.59
0.58
0.57
80
85
90
95
100
$m_{\mu^+\mu^-}$ (GeV/c^2)
Z line shape (arb. units)
linear fit 85-98 GeV: +0.23 %/GeV
toy Z: both muons accepted